

CS1231S Tutorial #1 Solutions

Tutorial questions only. Discussion questions excluded. Solutions follow the CS1231S proof-writing style with numbered steps and justifications where required.

1. If and only if / necessary and sufficient conditions

(a) p : I use the umbrella. q : It rains.

"I use the umbrella if it rains": $q \rightarrow p$

"I use the umbrella only if it rains": $p \rightarrow q$

(b) For $q \rightarrow p$: q is sufficient for p . p is necessary for q .

For $p \rightarrow q$: p is sufficient for q . q is necessary for p .

(c) p iff q : $(p \rightarrow q)$ and $(q \rightarrow p)$

Shorter form: $p \leftrightarrow q$

(d) "I use the umbrella" is both necessary and sufficient for "it rains".

2. De Morgan's law mistakes

(a) Incorrect: a and $\sim(b \text{ and } c)$ equivalent to a and $\sim b$ or $\sim c$

The negation only applies to the entire conjunction: $\sim(b \text{ and } c)$ equivalent to $\sim b$ or $\sim c$

Correct: a and $(\sim b \text{ or } \sim c)$

The omitted parentheses change the meaning.

(b) Incorrect: $\sim(x \text{ or } y) \text{ or } z$ equivalent to $\sim x$ and $\sim y$ or z

The correct application is: $\sim(x \text{ or } y)$ equivalent to $\sim x$ and $\sim y$

Therefore: $(\sim x \text{ and } \sim y) \text{ or } z$

3. Simplification

(a) $\sim a$ and $(\sim a \rightarrow (b \text{ and } a))$

$= \sim a$ and $(a \text{ or } (b \text{ and } a))$ (Implication law)

$= \sim a$ and $((a \text{ or } b) \text{ and } (a \text{ or } a))$ (Distributive law)

$= \sim a$ and $((a \text{ or } b) \text{ and } a)$ (Idempotent law)

$= \sim a$ and a (Absorption law)

$= \text{false}$ (Negation law)

(b) $(p \text{ or } \sim q) \rightarrow q$

$= \sim(p \text{ or } \sim q) \text{ or } q$ (Implication law)

$= (\sim p \text{ and } q) \text{ or } q$ (De Morgan's law)

$= q$ (Absorption law)

(c) $\sim(p \text{ or } \sim q) \text{ or } (\sim p \text{ and } \sim q)$

$= (\sim p \text{ and } q) \text{ or } (\sim p \text{ and } \sim q)$ (De Morgan's law)

$= \sim p \text{ and } (q \text{ or } \sim q)$ (Distributive law)

$= \sim p \text{ and true}$ (Negation law)

$= \sim p$ (Identity law)

(d) $(p \rightarrow q) \rightarrow r$

$= (\sim p \text{ or } q) \rightarrow r$ (Implication law)

$= \sim(\sim p \text{ or } q) \text{ or } r$ (Implication law)

$= (p \text{ and } \sim q) \text{ or } r$ (De Morgan and Double Negation laws)

4. Logical equivalence

Claim: $(p \rightarrow q) \rightarrow r$ is logically equivalent to $p \rightarrow (q \rightarrow r)$.

Proof by simplification:

$(p \rightarrow q) \rightarrow r$

$= \sim(\sim p \text{ or } q) \text{ or } r$ (Implication law)

$= (p \text{ and } \sim q) \text{ or } r$ (De Morgan and Double Negation laws)

For the other side:

$p \rightarrow (q \rightarrow r)$

$= \sim p \text{ or } (\sim q \text{ or } r)$ (Implication law)

$= \sim p \text{ or } \sim q \text{ or } r$

These are not equivalent.

Counterexample: Let $p = \text{false}$, $q = \text{true}$, $r = \text{false}$.

$(p \rightarrow q) \rightarrow r$: $p \rightarrow q$ is true, so $\text{true} \rightarrow \text{false}$ is false.

$p \rightarrow (q \rightarrow r)$: $q \rightarrow r$ is false, so false $\rightarrow$ false is true.

Therefore they are not logically equivalent.

5. Conditional statement transformations

Original: If $12x - 7 = 29$, then $x = 3$.

Negation: $12x - 7 = 29$ and $x \neq 3$.

Contrapositive: If $x \neq 3$, then $12x - 7 \neq 29$.

Converse: If $x = 3$, then $12x - 7 = 29$.

Inverse: If $12x - 7 \neq 29$, then $x \neq 3$.

Original conditional: True. Solving: $12x - 7 = 29$ $12x = 36$ $x = 3$.

Converse: True.

If $x = 3$: $12(3) - 7 = 29$.

The converse can be true while the inverse is false because the converse is logically equivalent to the inverse.

6. Alternative definitions of implication

For each alternative, provide a counterexample to transitivity.

Alternative 1: Choose $p = \text{false}$, $q = \text{true}$, $r = \text{false}$.

$p \rightarrow_a q$ is false. Therefore the premises fail, so find another assignment where: $(p \rightarrow_a q)$ and $(q \rightarrow_a r)$ are true but $(p \rightarrow_a r)$ is false.

$p = \text{true}$, $q = \text{true}$, $r = \text{false}$: $p \rightarrow_a q = \text{true}$, $q \rightarrow_a r = \text{false}$.

Hence not sufficient.

Alternative 2: $p = \text{false}$, $q = \text{true}$, $r = \text{false}$: $p \rightarrow_b q = \text{true}$, $q \rightarrow_b r = \text{false}$.

Alternative 3: $p = \text{false}$, $q = \text{false}$, $r = \text{true}$: $p \rightarrow_c q = \text{true}$, $q \rightarrow_c r = \text{true}$, but $p \rightarrow_c r = \text{false}$.

Therefore transitivity fails for the alternative definitions.

7. Validity of arguments

(a) p and q therefore q

Valid. Rule: Specialization.

(b) $p \rightarrow q \sim p$ therefore $\sim q$

Invalid. Inverse error.

(c) $p \rightarrow q$ q therefore p

Invalid. Converse error.

(d) $p \rightarrow r$ $q \rightarrow r$ therefore $(p \text{ or } q) \rightarrow r$

Valid. Rule: Proof by division into cases.

8. Arguments

(a) Premises: $p \text{ or } (q \text{ and } r)$ $\sim p$

Critical row requires: $p = \text{false}$. Then q and r must both be true.

Therefore: $p = \text{false}$, $q = \text{true}$, $r = \text{true}$.

Argument is valid.

(b) Counterexample: $p = \text{false}$, $q = \text{false}$, $r = \text{false}$.

$p \text{ or } (q \text{ and } r)$ is true? No, so use: $p = \text{true}$, $q = \text{false}$, $r = \text{false}$.

First premise true. $\sim(p \text{ and } q)$ true. Conclusion r false.

Invalid.

(c) Let: $p = \text{I go to the beach}$. $q = \text{I take shades}$. $r = \text{I take sunscreen}$.

Premises: $p \rightarrow (q \text{ or } r)$ q and $\sim r$

Conclusion: p

Critical row: $p = \text{false}$, $q = \text{true}$, $r = \text{false}$.

Invalid.

(d) Let: $p = \text{buy new goat}$. $q = \text{buy used Yugo}$. $r = \text{need loan}$.

Premises: $p \text{ or } q$ $(p \text{ and } q) \rightarrow r$ q and $\sim r$

From q and $\sim r$, p cannot be true because p and q would imply r .

Valid. Rule: Elimination.

9. Knights and Knaves

Proof (by contradiction).

1. Suppose C and D are both knights. 1.1 What C says is true. (definition of knight) 1.2 Therefore D is a knave. (C says D is a knave) 1.3 What D says is true. (definition of knight) 1.4 Therefore C is a knave. (D says C is a knave) 1.5 C is not a knight. (every native is knight or knave, but not both) 1.6 Contradiction.

2. Therefore C and D are not both knights.

3. Suppose C and D are both knaves. 3.1 What C says is false. (definition of knave) 3.2 Therefore D is not a knave. 3.3 Contradiction.

4. Therefore C and D are not both knaves.

5. Since every native is either a knight or a knave, but not both: Conclusion: There is exactly one knight and one knave.

10. Product of odd integers

Proof.

Let x and y be arbitrary odd integers.

1. $x = 2m + 1$ for some m in $\mathbb{Z}$. (definition of odd)

2. $y = 2n + 1$ for some n in $\mathbb{Z}$. (definition of odd)

3. $xy = (2m+1)(2n+1)$. (substitution)

4. $xy = 4mn + 2m + 2n + 1$. (basic algebra)

5. $xy = 2(2mn+m+n)+1$. (basic algebra)

6. Let $j = 2mn+m+n$.

7. j is an integer. (closure of integers under addition and multiplication)

8. $xy = 2j+1$.

9. Therefore xy is odd. (definition of odd)

Hence the product of any two odd integers is an odd integer.

11. Odd square proof

(a) Smart's proof is incomplete.

It only proves: If n^2 is odd, then n is odd.

It does not prove: If n is odd, then n^2 is odd.

Therefore it does not prove an iff statement.

It is also not a proof by contradiction because the negation of the required statement was not assumed and no contradiction was derived.

(b)

Proof.

1. Suppose n is odd.
2. $n = 2r+1$ for some r in $\mathbb{Z}$. (definition of odd)
3. $n^2 = (2r+1)^2$. (substitution)
4. $n^2 = 4r^2+4r+1$. (basic algebra)
5. $n^2 = 2(2r^2+2r)+1$. (basic algebra)
6. Let $q = 2r^2+2r$.
7. q is an integer. (closure of integers under addition and multiplication)
8. $n^2 = 2q+1$.
9. Therefore n^2 is odd. (definition of odd)

Together with the other direction: n^2 is odd iff n is odd.